

14.7 NeuroPilot: A 28nm, 69.4fJ/node and 0.22ns/node, 32×32 Mimetic-Path-Searching CIM-Macro with Dynamic-Logic Pilot PE and Dual-Direction Searching

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Autonomous micro-robots, equipped with AI for urban navigation, are being deployed for various applications, such as package delivery and surveillance. These intelligent machines require efficient path-finding algorithms to navigate complex city environments effectively. Similarly, other domains like VLSI routing, urban path planning, and wavefront propagation also demand optimal navigation solutions, as illustrated in Fig. 14.7.1. While previous work has typically relied on digital accelerators to address navigation challenges [3-5], mimetic-path CIM architectures, which incorporate SRAM within their processing-element (PE) arrays, have demonstrated superior performance, power efficiency, and area utilization (PPA). However, existing mimetic-CIM solutions face several challenges: (1) single-direction searching (SDS) inefficiently utilizes half of the available search time; (2) DFF-based temporary storage suffers from poor PPA, while dynamic-logic based alternatives are prone to leakage issues; and (3) limited processing windows can lead to task failures in large-scale maps, but previous attempts to address these issues, such as map direct transfer and splitting, have their own limitations. This paper presents a 28nm 32×32 mimetic-path NeuroPilot CIM macro with (1) dual-direction searching (DDS) to accelerate the path searching; (2) a dynamic-logic pilot PE with an outspread pre-charge (OPC) circuit, achieving a superior PPA while reducing leakage; and (3) a novel 3-step coarse/fine-path (TsCFP) flow enabling efficient navigation of large maps. The proposed chip achieves a search rate of 3670M nodes/s, 213.6μm² PE area, and a 69.4fJ node energy consumption.

Figure 14.7.2 illustrates the overall structure and operation of the proposed pilot CIM: comprising of a 32×32 pilot PE array, WL and BL drivers, I/Os, column and row decoders, an OPC for sequential pre charging, and the main controller. Each pilot PE contains 14 SRAM cells and a PE core connected to 8 adjacent PEs. The proposed dual-direction searching (DDS) shows significant improvement over single-direction searching (SDS), as shown in an ISSCC-like map example in Fig. 14.7.2. DDS initiates paths, starting at both the START and END nodes, meeting at a junction PE; thereby, achieving a 1.13× lower power consumption and a 2.1× faster processing time for the example map. It also achieves a 1.28× power and 2.0× processing improvements for a 32×32 map without block PEs. The NeuroPilot CIM workflow involves map preprocessing, chip controller decision-making for coarse/fine mapping or direct CIM macro processing based on map size. The CIM macro operates in three stages: (1) a memory-write stage, where the map data, including BLOCK (unreachable cells), START and END points, and the traffic-jam level (TJL), are written to the PE SRAM cells; (2) a path-pilot stage, which initiates a path search that runs until convergence, while also storing direction information in PE SRAM cells; and (3) a memory-read stage, where the final path is determined by tracing from the junction PE.

Figure 14.7.3 shows a detailed schematic for the pilot PE, featuring 14b SRAM cells with two types of cells: SRAM_F for storing EN, STA, END, and TJL using a 6T cell and inverter-driver, and SRAM_R for direction data using a 6T cell, local reset, and a inverter-writer. The PE comprises of 8 direction transmitters and a center core which processes the first input signal, IN[2:0], to arrive from adjacent PEs, determines its origin, START or END, and halts further input processing. The three input-signal bits IN[0] – IN[2] represent transmitting signal arrive, signal originates at START, and signal originates at END, respectively. Each transmitter handles its corresponding inputs, transmits the opposite outputs, and stops receiving other inputs. Take N as an example, N transmitter handles inputs from NW, N and NE, transmits S outputs. Redundancy in design, via double dynamic logic and extra input signals, mitigate potential PVT variation errors. The TJL circuit uses a delay tree to simulate path speed. For example, if this node has a traffic jam, then the EMA [2:0] will be 111, which means this node has the slowest speed. When two signals, originating at both START and END are received, the PE generates a junction signal to deactivate the PE and produce row and column signals to mark the junction's position. The dynamic logic design chooses MOS capacitance rather than MOM capacitance for its high density.

Figure 14.7.4 illustrates the proposed macro's outspread pre-charging (OPC) flow. Leveraging the PE array's logic propagation delay the OPC selectively pre-charges PEs, starting with the START and END rows and progressively transmits the pre-charging signal to adjacent (up and down) rows. This approach ends the propagation path upon START and END convergence, where the pre-charge signal is slightly faster than PE array signals. The OPC implementation results in significant improvements: a 1.49× hold-time reduction, leading to a 1.35× lower dynamic-logic capacity requirement and a 1.32× smaller area. The design incorporates feedback row and column signals, connected separately, which are processed by column and row encoders to determine the converging row and column. These signals pinpoint the junction; the proposed macro reads the corresponding SRAM and IO transmit directional data to the PC for further processing.

Figure 14.7.5 illustrates the proposed 3-step coarse/fine-path (TsCFP) flow that addresses the limitations of previous large map-conversion methods. This approach overcomes the vagueness issues of direct-map transfers and local-optima problems of split-map transfers. The process unfolds in three stages: (1) generation of the smallest clear coarse map by iteratively adjusting the resolution and scoring based on a path-ratio comparison between the original and converted maps; (2) creation of a coarse map's shortest path by splitting it into 32×32 segments for piloting, via SDS or DDS, with subsequent maps determined by previous END points and the original map's general direction; and (3) fine-map shortest-path generation by splitting the original map along the shortest coarse path and processing segments sequentially. When applied to an example based on a map of San Francisco the proposed flow achieves a 373.82ns time to solution (TTS), a 512.5pJ energy consumption, and a 1279.8L(unit length) path-length solution.

Figure 14.7.6 presents the measured test-chip results. The test chip is fabricated using a 28nm technology using a 32×32 NeuroPilot CIM macro. Dynamic-logic fading analysis demonstrates that the proposed macro is suitable for tasks up to 360L (84ns), encompassing both real-world scenarios (50-100L) and complex maze-like maps (>100L). The proposed macro occupies a 14.5 × 14.7 = 213.6μm² PE area, consumes of 69.4fJ/node, and achieves a search rate of 3670M nodes/s. In comparison to prior path-searching work [1-5], this work improves the search-rate / energy-per-node / PE-size FoM by 31.49 - 2.148e7×. Figure 14.7.7 shows the die photograph and a design summary table; also shown are short-distance and long-distance search examples.

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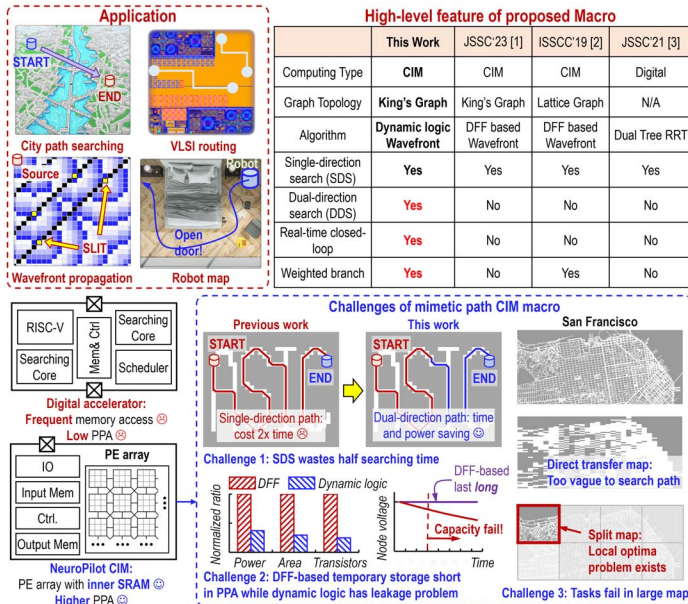


Figure 14.7.1: Motivation and challenges of mimetic path CIMs.

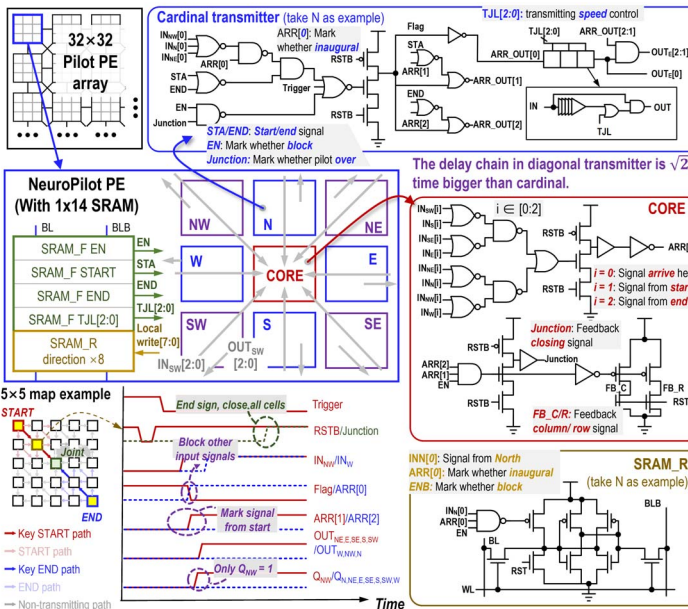


Figure 14.7.3: Detailed pilot PE circuit and operational waveforms.

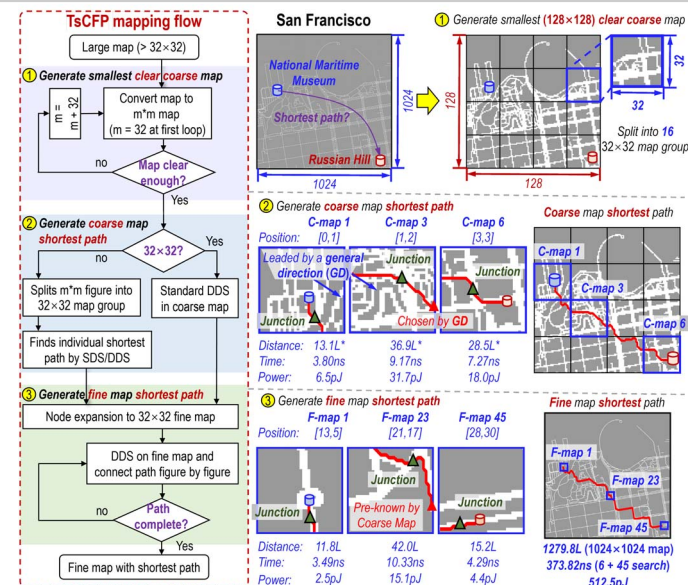


Figure 14.7.5: 3-step coarse/fine-path (TsCFP) mapping flow and an example based on a 1024x1024 map of San Francisco.

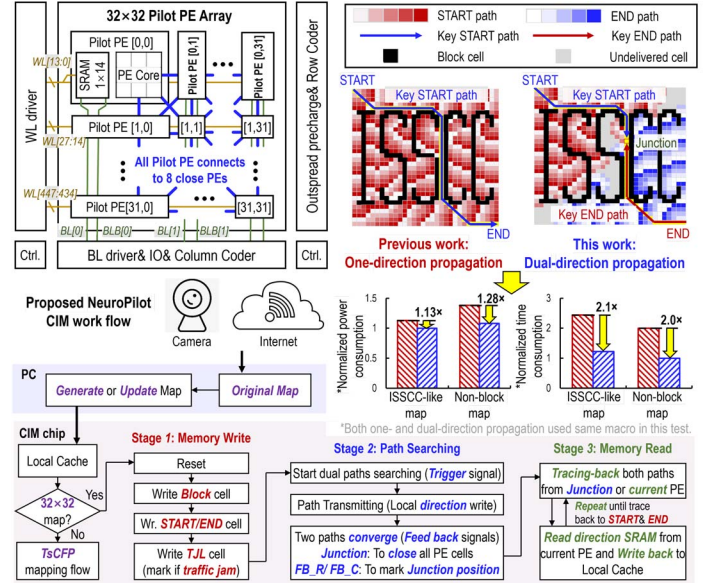


Figure 14.7.2: Proposed NeuroPilot CIM structure, dual-direction search (DDS) propagation and NeuroPilot CIM flow.

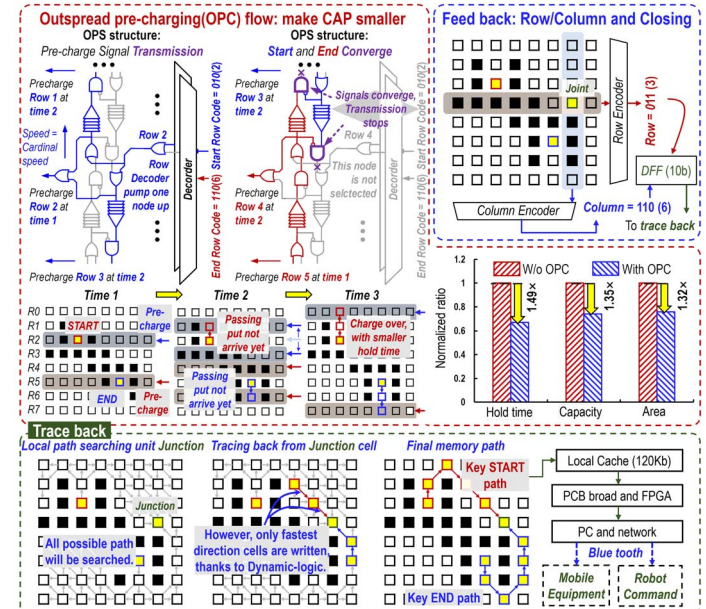


Figure 14.7.4: Outspread pre-charging, row/column feedback, and trace-back flow.

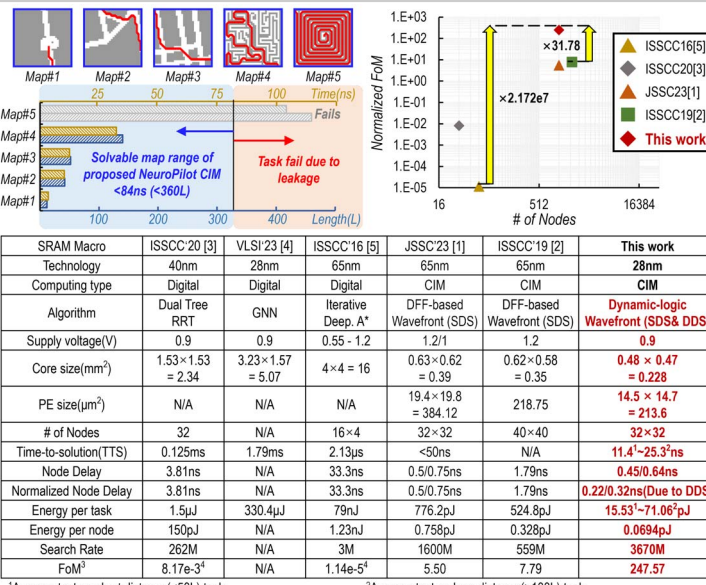


Figure 14.7.6: Measurement results and comparison table.

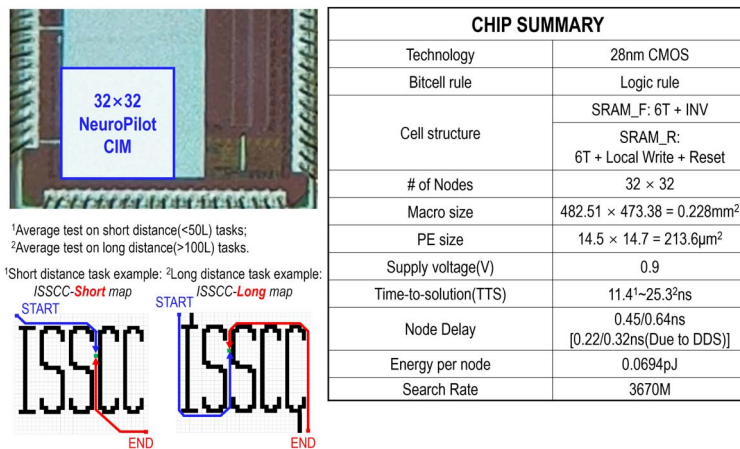


Figure 14.7.7: Die photograph and key-metric summary.

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